Data Query Language

Data Query Language (DQL) allows us to communicate with databases, facilitating precise **extraction and manipulation of data.**

SELECT

Select specific or all columns from a specified table. SELECT* selects all columns from a table. It is good to use the LIMIT statement, especially for large tables.

SELECT ALL

```
SELECT
    *
FROM
    table_name

LIMIT
    100;
```

SELECT Columns

```
SELECT
    column1,
    column2,
    column3

FROM
    table_name

LIMIT
    100;
```

SELECT DISTINCT

SELECT DISTINCT eliminates rows where data are **duplicated.** Like **SELECT**, we can check if data in specific column(s) or all columns are duplicated, and remove them.

```
SELECT DISTINCT
    column1,
    column2,
    column3

FROM
    table_name,
```

SELECT WHERE

The **WHERE** clause is used to **filter** records. It is used to extract only those records that fulfil a specified condition.

```
SELECT
    column1,
    column2,
    column3

FROM
    table_name,

WHERE
    column = value;
```

Operators

Operators are used to perform specific operations on one or more operands (values or expressions). Operators are categorised into several types: comparison operators, logical operators, and others like the LIKE operator and WILDCARD.

Comparison Operators

Comparison operators like **greater than** (>), **less than** (<), **equal to** (=), **not equal to** (<> **or** !=), **more than or equal to** (<=) are used with WHERE to find rows where the outcome of the comparison between a column value and a specified value is TRUE.

Greater Than

```
SELECT
    *
FROM
    table_name

LIMIT
    column1 > value;
```

Less Than

```
SELECT
    *
FROM
    table_name

LIMIT
    column1 < value;
```

Equal To

```
SELECT
    *
FROM
    table_name

WHERE
    column1 = value;
```

Not Equal To

```
SELECT
    *
FROM
    table_name

WHERE
    column1 != value;
```

Logical Operators

Logical operators are used to **combine or negate conditions**, determining the overall truth of a condition or a set of conditions.

AND

The **AND** operator combines multiple conditions in a query requiring **all** conditions to be met for a row to be included in the result set.

```
SELECT
    column1,
    column2,
    column3

FROM
    table_name

LIMIT
    condition1
    AND condition2;
```

OR

The **OR** operator displays a record in a query where **either** condition separated by the **OR** is **TRUE.**

```
SELECT
    column1,
    column2,
    column3

FROM
    table_name

LIMIT
    condition1
    AND condition2;
```

IN and NOT IN

The IN and NOT IN operators filter data based on a list of specificed values, allowing for inclusion or exclusion of rows that either do or do not match the provided set of values.

IN

```
SELECT
    column1,
    column2,
    column3

FROM
    table_name

WHERE
    column1

IN
    (
    value1,
    value2,
    value3);
```

NOT

```
SELECT
    column1,
    column2,
    column3

FROM
    table_name

WHERE
    column1

NOT IN
    (
    value1,
    value2,
    value3);
```

BETWEEN

The BETWEEN operator selects rows within a **specified range**, inclusive of the boundaries, enabling efficient filtering of data based on a given range of values.

BETWEEN

```
SELECT
    column1,
    column2

FROM
    table_name

WHERE
    column1

BETWEEN lower_value AND
    upper_value;
```

NOT BETWEEN

```
SELECT
    column1,
    column2

FROM
    table_name

WHERE
    column1

NOT BETWEEN lower_value AND
    upper_value;
```

LIKE and WILDCARDS

The LIKE operator is used with **wildcard characters** such as ('%' and/or '_') to allow for **pattern-based searching.**

% before

```
SELECT
    column1,
    column2

FROM
    table_name

WHERE
    column1

LIKE
    "%XXXX";
```

% after

```
SELECT
    column1,
    column2

FROM
    table_name

WHERE
    column1

LIKE
    "XXXX%";
```

IS NOT NULL/IS NULL

A field with a **NULL** value is a field with **no value.** The IS NULL statement will select values in a column that contains **NULL** values. The IS NOT NULL statement will select values with no nulls in them.

Select values with IS NOT NULL

```
SELECT
    column1,
    column2

FROM
    table_name

WHERE
    column1 IS NOT NULL;
```

Select values with IS NULL

```
SELECT
    column1,
    column2

FROM
    table_name

WHERE
    column1 IS NULL;
```